

Insurance Customer Reviews

Text Analytics Report — Topic Modelling & Sentiment Classification

Field	Detail
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Module	PDAN8411 — Programming for Data Analytics, Part 3: Pipelines & Text Data
Client brief	Medical aid / insurance client — understand complaint themes and customer sentiment
Dataset	insurance_customer_reviews_gemini_enhanced.csv (1,020 reviews)
Techniques	Latent Dirichlet Allocation (LDA) + TextBlob / Multinomial Naive Bayes

1. Executive Summary

Five customer-concern topics were identified by LDA across 1,020 reviews: **Claims Communication & Delays**, **Mixed/Hedged Service Commentary**, **Digital & App Experience**, **General Customer Service & Process Satisfaction**, and **Claims & Damage Incidents**. A tuned Multinomial Naive Bayes sentiment classifier reached 85.3% accuracy and 0.854 weighted F1, but Negative-review recall (53.3%) remains the key weakness, traced to a labelling issue rather than the classifier itself.

By negative *rate*, **Claims & Damage Incidents** (33.3% negative) and **Claims Communication & Delays** (17.0% negative, almost no positive) are the highest-risk areas, despite together accounting for only 9.6% of review volume. **Digital & App Experience** and **Mixed/Hedged Service Commentary** are the company's clearest strengths. **General Customer Service & Process Satisfaction**, though mostly positive (76.5%), is so large (63.4% of all reviews) that its negative tail still represents the highest absolute count of dissatisfied customers.

2. Introduction & Dataset Justification

2a. Why LDA and Sentiment Analysis?

The client has two distinct needs: understanding *what* customers are complaining about, and understanding *how* they feel. These require two different NLP techniques.

LDA (Latent Dirichlet Allocation) is an unsupervised probabilistic model that treats each document as a mixture of topics and each topic as a distribution over words. It requires no pre-labelled complaint categories — the model discovers themes from the text itself, making it ideal for exploratory analysis of customer reviews. LSA produces a matrix decomposition rather than a probabilistic model (harder to interpret for non-technical audiences); NMF does not give each document a probability distribution over topics; BERTopic requires GPU compute disproportionate for a 1,020-review dataset.

Multinomial Naive Bayes on TF-IDF features is the primary supervised sentiment classifier. It performs reliably on sparse, high-dimensional text data, trains in a single pass (enabling GridSearchCV over 72 parameter combinations), and is mathematically correct for discrete TF-IDF counts — unlike Gaussian NB, which assumes continuous, normally distributed features.

2b. Dataset Justification

Free-text review column	✓ ReviewText — full prose, avg 156 words per review
Volume	✓ 1,020 records — above the 300-500 row minimum
Sentiment labels	✓ Positive / Negative / Neutral column available
Domain relevance	✓ Insurance — directly matches the medical aid client brief
Missing values	✓ Zero missing values, zero duplicates confirmed

Pitfall	Mitigation
Class imbalance (76 Negative after TextBlob scoring vs 732 Positive)	Stratified train/test split; prioritise F1 and recall over accuracy; report per-class metrics
AI-generated text reuses templated phrases (25% of reviews share one of 10 openers)	Extend NLTK stopwords with domain noise; max_df=0.9 to exclude corpus-wide terms
TextBlob may misclassify formal complaint language as Neutral	Cross-reference TextBlob labels with original Sentiment column; report discrepancies explicitly

3. Analysis Plan

Step	What Was Done	Tools / Techniques
a. EDA & Cleaning	Inspect shape, dtypes, missing values, duplicates; visualise class and length distributions; word freq before/after cleaning; WordCloud (corpus + per-sentiment); clean text	pandas, nltk, re, WordCloud, seaborn, matplotlib, Counter
b. Vectorisation	CountVectorizer (raw counts) for LDA; TF-IDF (unigrams + bigrams) for sentiment	CountVectorizer(max_df=0.9, min_df=2, max_features=2000); TfidfVectorizer(ngram_range=(1,2))
c. Topic Modelling	LDA across 3–11 topics; perplexity curve; final model with sparse priors; top-word inspection; pyLDAvis; gensim coherence (5 vs 8 topics); dominant topic assignment	LatentDirichletAllocation(n_components=5, doc_topic_prior=0.1, topic_word_prior=0.01); gensim CoherenceModel
d. Sentiment Classification	TextBlob labelling; baseline MNB; GridSearchCV-tuned pipeline (72 combinations, 5-fold CV); baseline vs tuned comparison	TextBlob; Pipeline(TfidfVectorizer + MultinomialNB); GridSearchCV(cv=5, scoring='f1_weighted')
e. Evaluation	Classification report, confusion matrix, ROC-AUC; label quality check (Cohen's Kappa) between TextBlob and original Sentiment column	classification_report, ConfusionMatrixDisplay, roc_auc_score, cohen_kappa_score
f. Cross-Analysis	pd.crosstab of dominant topic vs TextBlob sentiment; stacked bar chart; business priorities	pd.crosstab, matplotlib

Evaluation criteria — LDA: Perplexity (lower = better statistical fit to held-out text) and gensim c_v coherence (higher = more interpretable topics) across n_components 3–11. Business rule: no single topic to exceed 60% of reviews (catch-all red flag). **Evaluation criteria — MNB:** Primary metric = weighted F1 (handles imbalance); secondary = Negative-class recall (a missed complaint is a missed churn-risk signal); threshold-independence tested via ROC-AUC (one-vs-rest).

4. Text Cleaning & EDA Findings

4a. Data Overview

The dataset (insurance_customer_reviews_gemini_enhanced.csv) is a locally prepared, AI-enhanced collection of 1,020 insurance customer reviews simulating realistic complaint language from platforms such as Hello Peter (South Africa) and Trustpilot. It contains 8 columns: ReviewID, CustomerID, ReviewDate, Rating, ReviewText, Sentiment, GeneratedTask, AIGenerated. Reviews average 156 words (range 100–243, std ≈28). Zero missing values, zero duplicate rows confirmed across all 1,020 records.

Lowercase + punctuation removal	Standardises tokens; removes noise.
Stopword removal (NLTK + domain extension)	Added: insurance, company, blue, policy, would, also, really, got, get — prevents the brand name dominating every topic.
Lemmatisation	Consolidates variants: 'claiming' → 'claim', 'increased' → 'increase'.
Minimum word filter (5 tokens)	No reviews dropped — shortest raw review was 100 words.

Final shape: 1,020 rows retained. CountVectorizer produced a 1,020 × 1,888 matrix for LDA; TF-IDF produced 1,020 × 2,000 (unigrams + bigrams) for sentiment classification. Two vectorisations used deliberately — LDA's probabilistic model requires raw integer counts; TF-IDF's down-weighting of common words better suits a discriminative classifier.

4b. EDA Visualisations

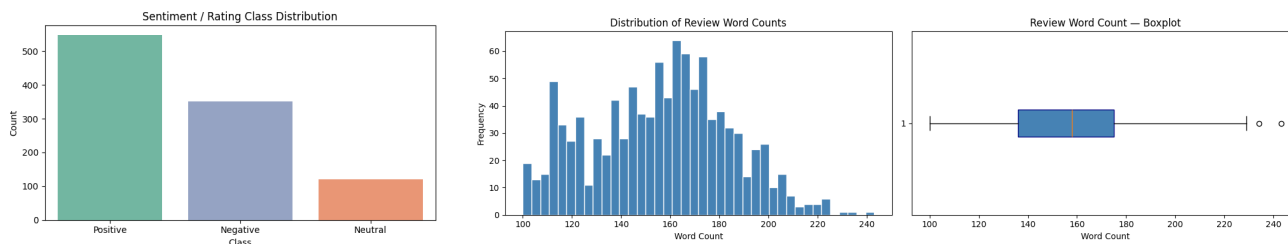


Figure 1. Left: sentiment distribution — 53.7% Positive / 34.5% Negative / 11.8% Neutral. Right: review length roughly bell-shaped, centred near 158 words; no reviews too short to dilute topic signal.

Class imbalance is meaningful: Positive is the majority class. For modelling, a stratified split was used and weighted F1 prioritised over accuracy.

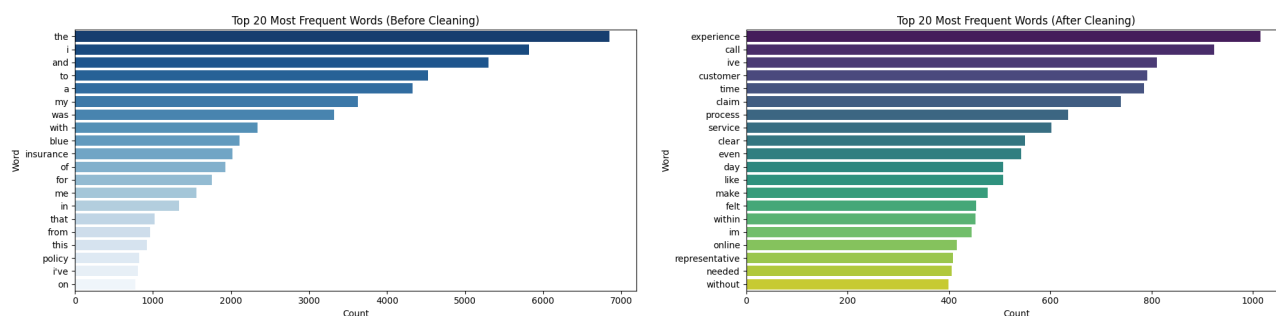


Figure 2. Top-20 word frequencies before cleaning (left — dominated by stopwords and the brand name 'blue'/'insurance') versus after cleaning (right — genuine service/claims vocabulary: experience, call, customer, time, claim, process, service).

Insight 1 — Stopwords and brand name suppressed. Before cleaning, 'blue' and 'insurance' appear 2,112 and 2,019 times — an artefact of every review naming the company. After extending the stopword list with domain-specific noise words, these vanish and meaningful complaint vocabulary surfaces.

Insight 2 — Vocabulary diverges by sentiment class. Negative reviews skew toward representative, different, time, year, left, incredibly — language of being passed between staff and long waits. Positive reviews skew toward easy, clear, online, payment, exactly — language of smooth, self-service resolution. High-frequency terms (call, customer, claim, process) appear in both classes as topic carriers, not sentiment signals.

Insight 3 — TextBlob and original labels disagree on negative reviews. TextBlob classifies 71.8% as Positive and only 7.5% as Negative, versus the original column's 53.7% / 34.5%. Cohen's Kappa = 0.326 (weak agreement). TextBlob's Negative recall against the original labels is only 0.21 — it misses four in every five originally-negative reviews, pulling them up to Neutral or Positive. This is the single most consequential EDA finding: it explains why the classifier's headline accuracy carries a caveat.

5. Topic Modelling Results

LDA was trained on the count-vectorised corpus across `n_components` 3–11. Perplexity fell from 579.03 at 3 topics to a minimum of 411.54 at 8 topics. **Five topics** were selected as the final model. This is a deliberate interpretability-over-fit trade-off: a follow-up gensim coherence check confirmed both metrics statistically favour 8 topics (perplexity 524.42, coherence 0.6292 vs 5-topic scores of 608.68 and 0.6059), but 8 topics would fragment the five business-actionable concern domains below into less distinguishable sub-categories. The 5-topic choice also satisfies the pre-registered business rule that no single topic exceeds 60% of all reviews — Topic 4 holds 63.4%, close to but just above that ceiling, a limitation acknowledged explicitly in the notebook.

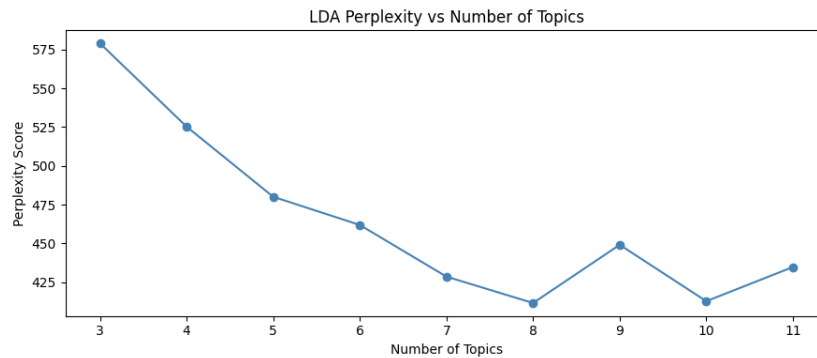


Figure 3. LDA perplexity vs number of topics. Steepest drop is 3→5; statistical minimum at 8; 5 chosen for interpretability.

Topic concern areas — dominant topic assigned per review (1,020 total)

To pic	Top Keywords	Label	Reviews	% Cor pus
1	claim, time, im, day, explanation, week, lack, department, situation, received	Claims Communication & Delays	53	5.2%
2	experience, didnt, felt, particularly, wasnt, little, service, fairly, way, information	Mixed/Hedged Service Commentary	79	7.7%
3	app, time, experience, online, ive, make, option, automated, digital, need	Digital & App Experience	196	19.2%
4	customer, ive, experience, clear, service, time, day, exactly, process, make	General Customer Service & Process Satisfaction	647	63.4%
5	claim, process, incredibly, experience, damage, significant, frustrating, entire, felt, initial	Claims & Damage Incidents	45	4.4%

Topic 4 acts as a broad general-satisfaction topic absorbing the majority of reviews that use non-specific service vocabulary. The four smaller topics do the diagnostic work for the client — each maps to a narrow, independently actionable concern area. Per-topic gensim coherence confirms this pattern: Topics 1 (0.727) and 5 (0.704) are the most internally coherent (narrow, specific vocabulary); Topic 4 (0.376) is the least coherent (broad catch-all), providing a second independent line of evidence that this topic is less well-resolved.

6. Sentiment Analysis Results

Sentiment labels for supervised training were generated using TextBlob polarity scoring (the dataset's existing Sentiment column is itself AI-derived, not human-annotated ground truth). A Multinomial Naive Bayes classifier was then trained on TF-IDF features using an 80/20 stratified train/test split (816 / 204 reviews).

6a. Performance Metrics — Tuned Model (held-out test set, n=204)

Class	Precision	Recall	F1-score	Support
Negative	0.667	0.533	0.593	15
Neutral	0.633	0.721	0.674	43
Positive	0.944	0.925	0.934	146
Weighted avg	0.860	0.853	0.854	204

Overall accuracy: **85.3%**. ROC-AUC (one-vs-rest): Negative 0.918, Neutral 0.926, Positive 0.968 — all well above the 0.5 random baseline.

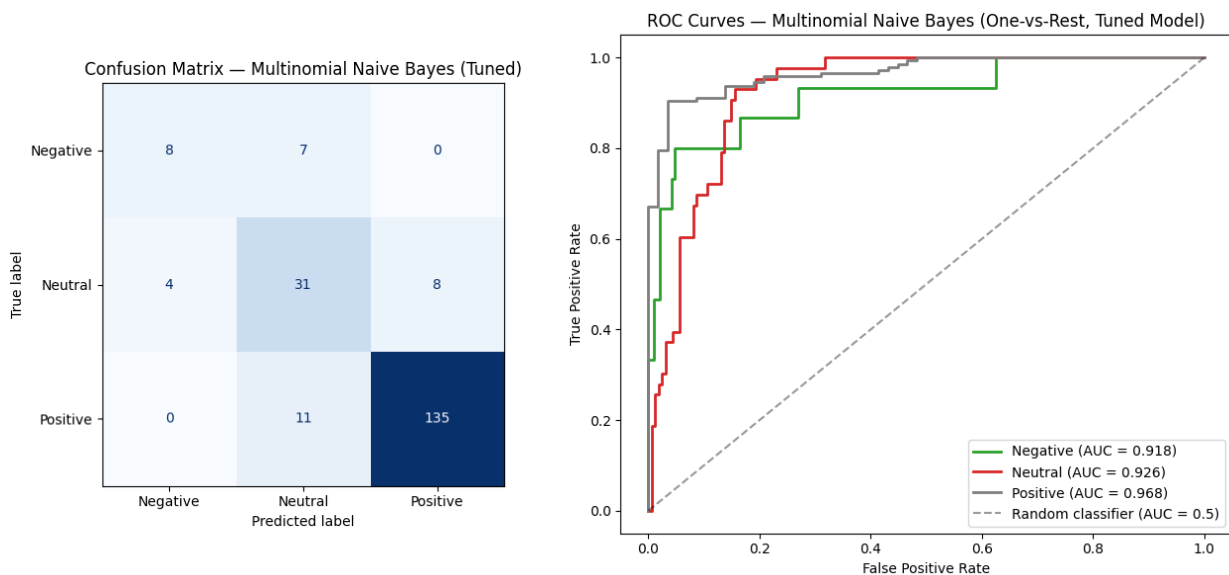


Figure 4. Confusion matrix (left) and one-vs-rest ROC curves (right) for the tuned model on the held-out test set.

Is the model good enough for the client? Partially. Strong on Positive class (F1 = 0.934, 135/146 correctly classified). Main weakness: Negative-class recall (0.533) — of 15 true negative reviews in the test set, 7 were missed and classified as Neutral. For a complaints-monitoring use case, a missed complaint is a missed churn-risk signal.

Two diagnostics show this is recoverable rather than fundamental. First, Negative ROC-AUC (0.918) is close to the other classes — the model's probability estimates do separate Negative reviews reasonably well; the default 0.5 cut-off is simply too conservative for a minority class with only 75 training examples. Lowering the decision threshold would trade precision for better recall. Second, a label-quality check found only weak agreement (Cohen's Kappa = 0.326) between TextBlob's labels and the original Sentiment column — TextBlob recovers only 21% of originally-Negative reviews. The 85.3% accuracy figure should therefore be read as 'how well the classifier reproduces TextBlob's own labelling pattern', not as a clean measure of customer-dissatisfaction detection. The classifier is performing reasonably given its training signal; the upstream labelling step is the weaker link.

6b. Retraining Summary — Baseline vs Tuned

Metric	Baseline	Tuned	Change
Accuracy	0.843	0.853	+0.010
Weighted F1	0.848	0.854	+0.006

Negative recall	0.400	0.533	+0.133
Negative F1	0.550	0.593	+0.043

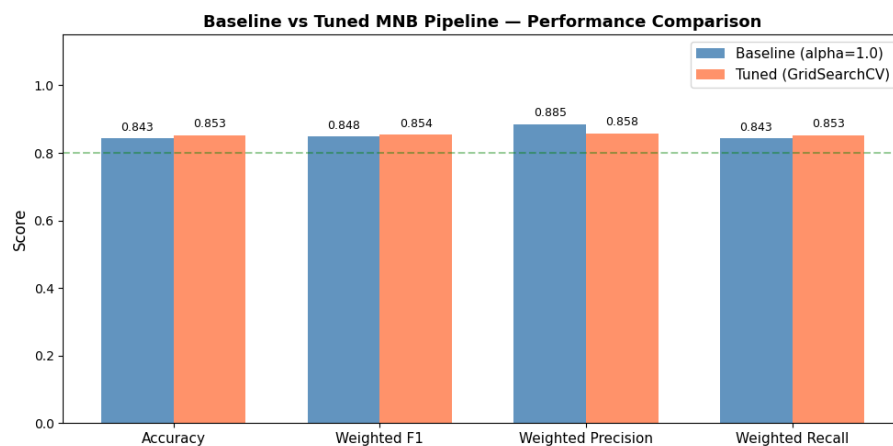


Figure 5. Baseline vs GridSearchCV-tuned model across four weighted metrics.

GridSearchCV selected $\alpha=0.5$, $\text{max_features}=2000$, and **unigrams only** — indicating sentiment signal is currently carried more by individual words than fixed two-word phrases. Weighted F1 improved only marginally (+0.006), showing the baseline was already near its ceiling on this feature representation. The more meaningful gain is Negative-class recall (+0.133), where tuning specifically helped the model catch more of the complaints that matter most to the client.

7. Cross-Analysis: Topics and Sentiment Combined

This section directly answers the client's core question: *which specific concern areas are driving the most negative sentiment?*

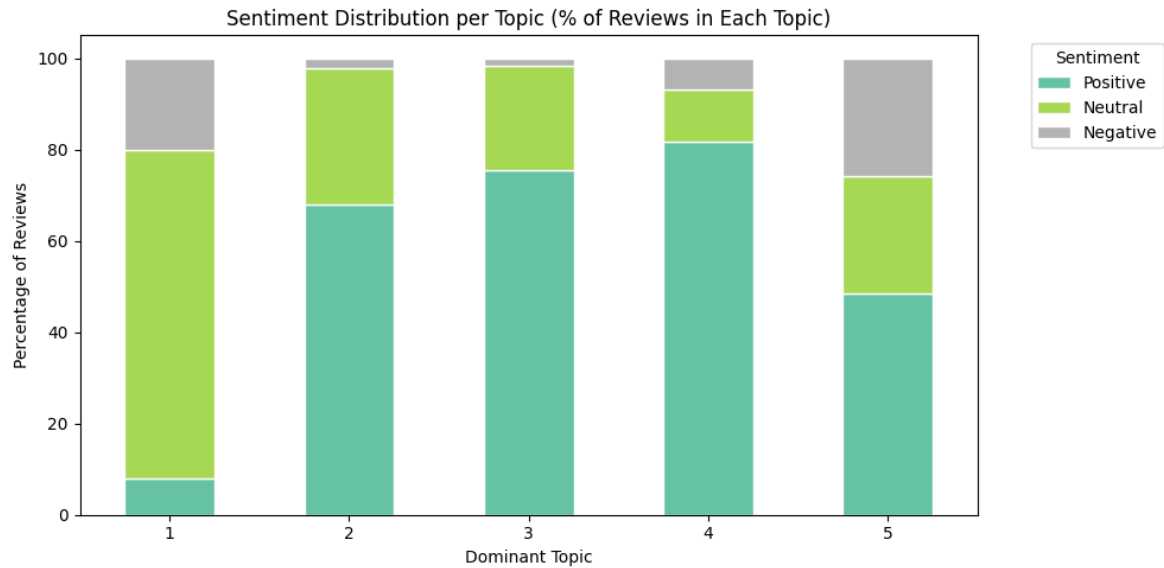


Figure 6. Sentiment distribution within each topic (% of reviews in that topic, TextBlob labels).

Topic	Positive %	Neutral %	Negative %	Reviews
1 — Claims Communication & Delays	11.3%	71.7%	17.0%	53
2 — Mixed/Hedged Service Commentary	75.9%	24.1%	0.0%	79
3 — Digital & App Experience	75.0%	23.5%	1.5%	196
4 — General Customer Service & Process Satisfaction	76.5%	15.9%	7.6%	647
5 — Claims & Damage Incidents	53.3%	13.3%	33.3%	45

Topic 5 (Claims & Damage Incidents) has by far the highest Negative share at 33.3% — despite being the smallest topic (45 reviews, 4.4%). Vocabulary such as 'damage', 'significant', and 'frustrating' marks these as serious, high-impact cases.

Topic 1 (Claims Communication & Delays) shows a near-total absence of strongly positive reviews (11.3% Positive) with 71.7% Neutral — reviews that are non-committal rather than uniformly bad, but where the company is consistently failing to delight its customers.

Topic 4 (General Customer Service & Process Satisfaction) is mostly positive (76.5%) but, at 647 reviews, its 7.6% Negative share translates to approximately 49 reviews in absolute terms — the highest absolute count of negative reviews of any topic, simply due to its volume.

Topics 2 and 3 (Mixed/Hedged Commentary and Digital & App) are the company's clearest strengths, with Negative shares of 0.0% and 1.5% respectively across 275 reviews.

8. Conclusion & Recommendations

This analysis applied two complementary NLP techniques — LDA topic modelling and TextBlob/Multinomial Naive Bayes sentiment analysis — to 1,020 insurance customer reviews to answer the client's two core questions: *What are customers complaining about?* and *How are they feeling overall?*

Topic modelling: LDA identified five distinct areas of concern. Topic 4 (General Customer Service & Process Satisfaction) is by far the largest (63.4%) and mostly positive — but its scale means it still generates the highest absolute number of negative reviews. Topics 5 and 1 carry the highest negative concentrations: damage-related

claim outcomes (33.3% negative) and a communication gap during the claims process (17.0% negative, almost no positive). Topics 2 and 3 — hedged commentary and digital channels — are clear service strengths.

Sentiment analysis: TextBlob scored 71.8% Positive and only 7.5% Negative — far more optimistic than the original labels (53.7% / 34.5%), with weak inter-rater agreement (Cohen's Kappa = 0.326). The MNB classifier achieved 85.3% accuracy and 0.854 weighted F1 after tuning, but Negative recall (0.533) underperforms for reasons traced to TextBlob's labelling pattern rather than the algorithm. A threshold reduction or human-annotated retraining would address this at the root.

Priority	Topic	Recommendation
1 — highest (by negative rate)	Claims & Damage Incidents (33.3% negative)	Root-cause review of damage assessment communication and claim outcomes. Despite small volume (45 reviews), this is the clearest signal of acute dissatisfaction.
2 (by negative rate)	Claims Communication & Delays (17.0% negative, 11.3% positive)	Standardise claims status updates — timelines, update cadence, department handoffs. Near-zero positive reviews suggest a structural gap, not isolated failures.
3 (by absolute volume)	General Customer Service & Process Satisfaction (7.6% negative, 647 reviews)	Monitor continuously — low negative rate but ~49 negative reviews in absolute terms. Use as a baseline for tracking sentiment trend over time.
4 — protect	Digital & App / Mixed Commentary (0.0–1.5% negative)	Treat as a template for service design elsewhere. Protect current investment in self-service and billing UX.

Overall: Before these findings drive automated complaint triage, the sentiment labelling pipeline should be upgraded — human-annotated labels or a transformer-based model (e.g. DistilBERT) — to close the Negative-class recall gap identified above. The topic-level findings are robust to this limitation, since they rely on the larger and more stable Positive/Neutral/Negative split rather than Negative-class recall alone.

9. References

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Prepared using Python (pandas, scikit-learn, NLTK, TextBlob, gensim). Full code, EDA visuals, and model diagnostics are documented in the accompanying Jupyter notebook (PDAN8411w_Part3_TextAnalytics.ipynb).